

Muon Life time Estimation

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Data Science and Analysis Project Report

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Abstract

The estimation of particle lifetimes is a fundamental problem in experimental particle physics and provides an important application of statistical inference. In this work, we analyze a dataset of muon decay times to estimate the mean lifetime of the muon. The decay process is modeled using an exponential probability distribution, and the lifetime parameter is determined using Maximum Likelihood Estimation (MLE).

To assess the validity of the model, goodness-of-fit tests such as the chi-square test and the Kolmogorov–Smirnov test are performed. The uncertainty in the estimated parameter is quantified using analytical confidence intervals as well as resampling techniques such as the bootstrap method. Additionally, model comparison is carried out using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) to verify the suitability of the exponential model against alternative distributions.

The results demonstrate that the exponential model provides a good description of the data, and the estimated muon lifetime is consistent with the accepted value within statistical uncertainty. This study highlights the role of statistical methods in parameter estimation, model validation, and uncertainty quantification in experimental physics.

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1 Introduction

Muon is a fundamental particle belonging to the lepton family. It is similar to the electron but has a much larger mass and is unstable in nature. Muons decay via weak interaction into an electron and neutrinos. Due to their well-understood decay mechanism, muons provide an excellent system for studying statistical methods in particle physics.

The decay of a muon is a random process and follows an exponential probability distribution. The characteristic parameter of this decay is the mean lifetime, denoted by τ , which is approximately $2.196 \mu s$. Estimating this lifetime from experimental data is a classical problem in data analysis and statistical inference.

In this project, we analyze a dataset of muon decay times and estimate the lifetime using statistical techniques. The primary goal is to determine the parameter τ and quantify the uncertainty associated with the estimate.

We employ the Frequentist framework for parameter estimation, including Maximum Likelihood Estimation (MLE), goodness-of-fit tests such as the chi-square test and Kolmogorov–Smirnov test, and uncertainty estimation methods such as confidence intervals and bootstrap resampling. Additionally, model comparison techniques such as Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used to validate the choice of the exponential model.

This study not only provides an estimate of the muon lifetime but also demonstrates how statistical tools are applied to validate models and quantify uncertainty in experimental physics.

2 Physical and Statistical Background

2.1 Muon Decay

Muon is an unstable lepton that decays via the weak interaction. The decay process is random and independent for each particle. The probability that a muon survives up to time t and then decays follows an exponential law.

The probability density function (PDF) for the decay time is given by:

$$p(t \mid \tau) = \frac{1}{\tau} e^{-t/\tau}, \quad t \geq 0 \quad (2.1)$$

where τ is the mean lifetime of the muon. The expected value and variance of the exponential distribution are:

$$\mathbb{E}[t] = \tau, \quad \text{Var}(t) = \tau^2 \quad (2.2)$$

The accepted value of the muon lifetime is approximately $2.196 \mu s$.

2.2 Statistical Inference

Statistical inference deals with estimating unknown parameters from observed data. In this work, the goal is to estimate the parameter τ from a set of observed decay times $\{t_1, t_2, \dots, t_n\}$.

2.3 Frequentist Approach

In the Frequentist framework, the parameter τ is treated as a fixed but unknown quantity, while the data are considered random variables. Estimation is performed using methods such as Maximum Likelihood Estimation (MLE).

For the exponential distribution, the likelihood function is:

$$L(\tau) = \prod_{i=1}^n \frac{1}{\tau} e^{-t_i/\tau} \quad (2.3)$$

Taking the logarithm:

$$\log L(\tau) = -n \log \tau - \frac{1}{\tau} \sum_{i=1}^n t_i \quad (2.4)$$

Maximizing this likelihood leads to the estimator:

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n t_i \quad (2.5)$$

Thus, the Maximum Likelihood Estimator of the lifetime is simply the sample mean.

2.4 Goodness of Fit

To evaluate how well the exponential model describes the data, goodness-of-fit tests are used.

2.4.1 Chi-Square Test

The chi-square test compares observed frequencies with expected frequencies from the model. The test statistic is given by:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad (2.6)$$

where O_i and E_i are the observed and expected counts, respectively.

2.4.2 Kolmogorov–Smirnov Test

The Kolmogorov–Smirnov (K–S) test compares the empirical distribution function of the data with the theoretical cumulative distribution function. The test statistic is defined as:

$$D = \sup_t |F_{\text{emp}}(t) - F_{\text{model}}(t)| \quad (2.7)$$

2.5 Uncertainty Estimation

Estimating the uncertainty in the parameter is essential.

2.5.1 Confidence Interval

For large sample sizes, the standard error of the estimator is:

$$\text{SE} = \frac{\tau}{\sqrt{n}} \quad (2.8)$$

A 95% confidence interval is given by:

$$\hat{\tau} \pm 1.96 \times \text{SE} \quad (2.9)$$

2.5.2 Bootstrap Method

The bootstrap method is a resampling technique used to estimate the distribution of an estimator. Multiple samples are drawn with replacement from the data, and the estimator is computed for each sample to obtain an empirical distribution.

2.6 Model Comparison

To compare different statistical models, information criteria are used.

2.6.1 Akaike Information Criterion (AIC)

$$\text{AIC} = 2k - 2 \log L \quad (2.10)$$

2.6.2 Bayesian Information Criterion (BIC)

$$\text{BIC} = k \log n - 2 \log L \quad (2.11)$$

where k is the number of parameters and n is the number of data points. Lower values of AIC and BIC indicate a better model.

3 Data Description and Exploratory Analysis

3.1 Dataset Description

The dataset consists of measured decay times of muons, recorded in units of microseconds (μs). Each entry in the dataset represents the time interval between the detection of a muon and its subsequent decay. The data is treated as a collection of independent and identically distributed observations.

The objective of this analysis is to study the statistical properties of the decay times and estimate the characteristic lifetime parameter of the underlying distribution.

3.2 Exploratory Data Analysis

To gain an initial understanding of the data, a histogram of the decay times is constructed. This provides a visual representation of the distribution and helps identify its general behavior.

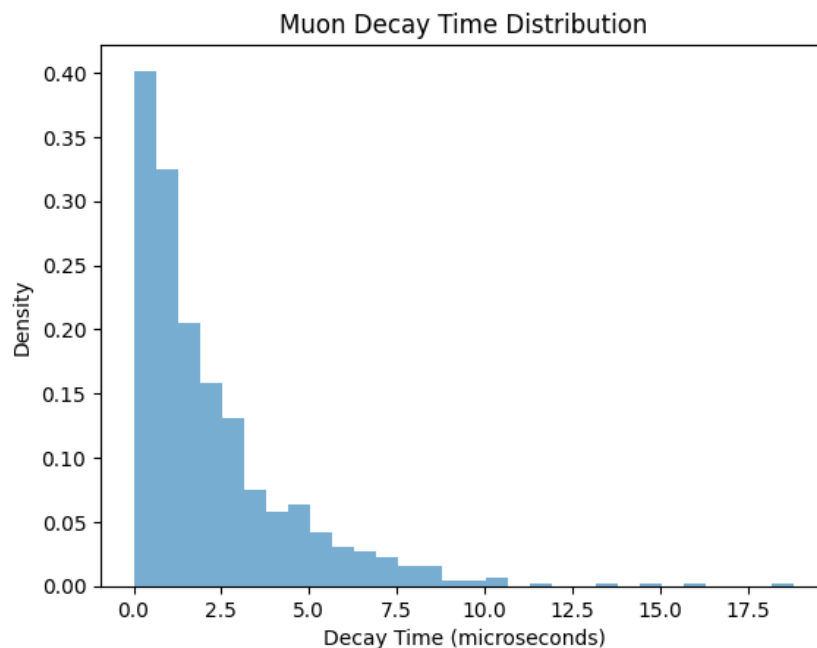


Figure 3.1. Histogram of muon decay times

From Figure 3.1, it can be observed that the distribution is highly skewed, with a larger number of events occurring at smaller decay times.

The probability density appears to decrease rapidly with increasing time, which is characteristic of an exponential decay process. This observation is consistent with the theoretical expectation for muon decay.

3.3 Initial Model Fitting

Based on the observed behavior, an exponential model is considered as a suitable candidate for describing the data. The parameter of the distribution is estimated using the sample mean, corresponding to the Maximum Likelihood Estimation (MLE) approach.

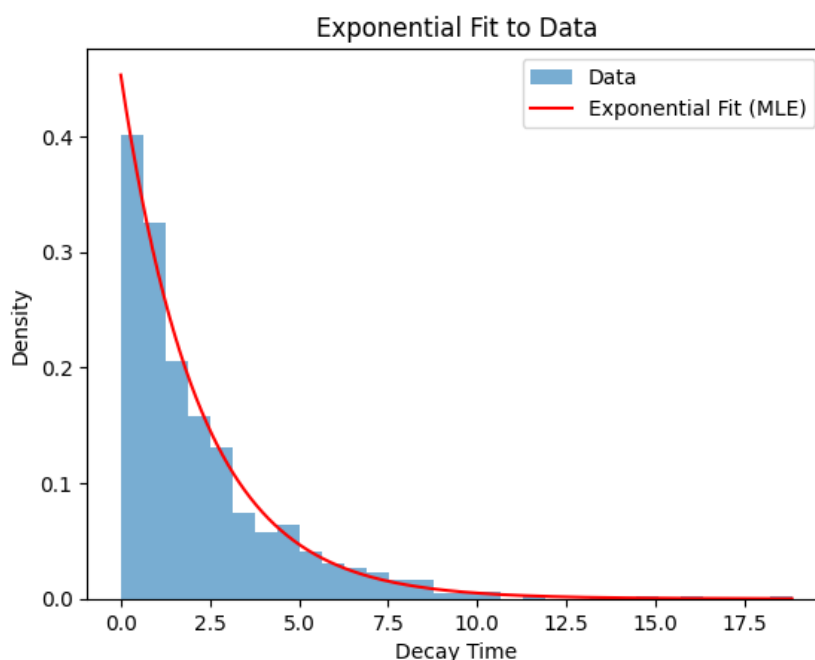


Figure 3.2. Exponential fit to the decay time distribution

The fitted exponential distribution is shown alongside the data histogram in Figure 3.2. The fitted curve provides a reasonable description of the observed distribution, capturing the overall trend of rapid decay at smaller times and a long tail at larger times.

3.4 Preliminary Observations

The exploratory analysis suggests that the exponential model is a plausible representation of the data. However, visual agreement alone is not sufficient to validate the model. A

more rigorous statistical analysis is required to quantify the quality of the fit and estimate the associated uncertainties.

In addition, resampling techniques are employed to study the variability of the estimated parameter. The bootstrap distribution of the estimated lifetime is shown in Figure 3.3.

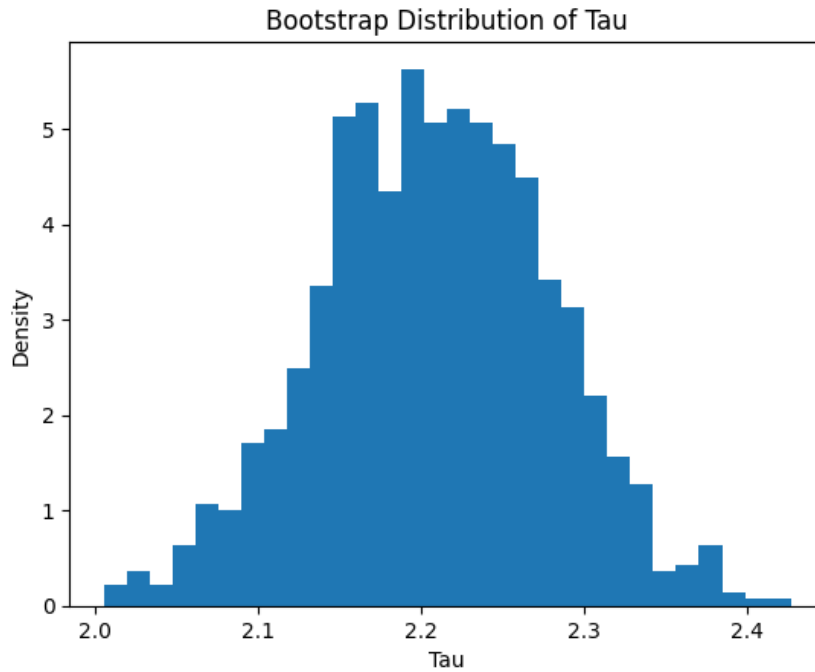


Figure 3.3. Bootstrap distribution of the estimated lifetime parameter

The bootstrap distribution provides insight into the stability and spread of the estimator, which will be analyzed quantitatively in subsequent sections.

3.5 Summary

The data exhibits behavior consistent with an exponential decay process. Initial parameter estimation using the sample mean provides a reasonable fit to the observed distribution. Further statistical analysis, including goodness-of-fit tests and uncertainty estimation, will be carried out in the following chapters to rigorously evaluate the model.

4 Frequentist Analysis

4.1 Maximum Likelihood Estimation

The decay times are modeled using an exponential probability distribution of the form:

$$p(t \mid \tau) = \frac{1}{\tau} e^{-t/\tau}, \quad t \geq 0 \quad (4.1)$$

Given a set of observed decay times $\{t_1, t_2, \dots, t_n\}$, the likelihood function is:

$$L(\tau) = \prod_{i=1}^n \frac{1}{\tau} e^{-t_i/\tau} \quad (4.2)$$

Taking the logarithm:

$$\log L(\tau) = -n \log \tau - \frac{1}{\tau} \sum_{i=1}^n t_i \quad (4.3)$$

Maximizing the likelihood leads to the estimator:

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n t_i \quad (4.4)$$

Thus, the Maximum Likelihood Estimator (MLE) of the lifetime is equal to the sample mean.

The estimated value obtained from the data is:

$$\hat{\tau} = 2.2078 \mu s \quad (4.5)$$

4.2 Goodness of Fit

4.2.1 Chi-Square Test

To evaluate the agreement between the observed data and the exponential model, the chi-square test is performed. The test statistic is defined as:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad (4.6)$$

The computed values are:

$$\chi^2 = 37.5442 \quad (4.7)$$

$$p\text{-value} = 0.1073 \quad (4.8)$$

The p-value is greater than the conventional significance level of 0.05, indicating that there is no statistically significant deviation between the observed data and the exponential model. Therefore, the model provides an adequate description of the data.

4.2.2 Kolmogorov–Smirnov Test

The Kolmogorov–Smirnov (K–S) test compares the empirical distribution of the data with the theoretical exponential distribution.

The test results are:

$$D = 0.0293 \quad (4.9)$$

$$p\text{-value} = 0.3496 \quad (4.10)$$

The relatively large p-value suggests that the null hypothesis cannot be rejected, further supporting the consistency of the data with the exponential model.

4.3 Uncertainty Estimation

4.3.1 Analytical Confidence Interval

The standard error of the estimator is given by:

$$SE = \frac{\hat{\tau}}{\sqrt{n}} \quad (4.11)$$

A 95% confidence interval is constructed as:

$$\hat{\tau} \pm 1.96 \times SE \quad (4.12)$$

The resulting interval is:

$$CI_{95\%} = [2.0709, 2.3446] \mu s \quad (4.13)$$

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4.3.2 Bootstrap Method

To further assess the variability of the estimator, the bootstrap method is employed. Multiple resampled datasets are generated, and the estimator is computed for each sample.

The bootstrap results are:

$$\text{Mean} = 2.2086 \quad (4.14)$$

$$\text{Standard deviation} = 0.0706 \quad (4.15)$$

The 95% bootstrap confidence interval is:

$$[2.0699, 2.3417] \mu s \quad (4.16)$$

The bootstrap distribution, shown in Figure 3.3, provides a visual representation of the uncertainty in the estimated parameter.

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4.4 Summary

The Maximum Likelihood Estimation yields a muon lifetime of $2.2078 \mu s$, which is consistent with the expected behavior of the decay process. Both the chi-square and Kolmogorov–Smirnov tests indicate that the exponential model provides a satisfactory description of the data.

The uncertainty in the estimate has been quantified using both analytical and bootstrap methods. The close agreement between the confidence interval and the bootstrap interval suggests that the estimation is robust and statistically reliable.

5 Model Comparison

5.1 Motivation

While the exponential distribution is theoretically motivated for muon decay, it is important to verify whether it provides a better description of the data compared to alternative models. Model comparison techniques allow for a quantitative assessment of competing hypotheses.

In this analysis, the exponential model is compared with a Gaussian model, which serves as a representative alternative distribution.

5.2 Candidate Models

5.2.1 Exponential Model

The exponential model is defined as:

$$p(t \mid \tau) = \frac{1}{\tau} e^{-t/\tau} \quad (5.1)$$

This model is physically motivated by the random and memoryless nature of muon decay.

5.2.2 Gaussian Model

The Gaussian model is given by:

$$p(t \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(t - \mu)^2}{2\sigma^2}\right) \quad (5.2)$$

Although not physically motivated for decay processes, it provides a useful comparison due to its widespread use in statistical modeling.

5.3 Information Criteria

To compare the models, the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used. These criteria penalize model complexity while rewarding goodness of fit.

$$\text{AIC} = 2k - 2 \log L \quad (5.3)$$

$$\text{BIC} = k \log n - 2 \log L \quad (5.4)$$

where k is the number of parameters and n is the number of observations.

5.4 Results

The computed values for the two models are:

$$\text{AIC}_{\text{exp}} = 3585.98 \quad (5.5)$$

$$\text{AIC}_{\text{gauss}} = 4476.15 \quad (5.6)$$

$$(5.7)$$

$$\text{BIC}_{\text{exp}} = 3590.89 \quad (5.8)$$

$$\text{BIC}_{\text{gauss}} = 4485.96 \quad (5.9)$$

5.5 Discussion

Both AIC and BIC values are significantly lower for the exponential model compared to the Gaussian model. Since lower values of these criteria indicate a better balance between goodness of fit and model complexity, the exponential model is strongly favored.

The difference in AIC between the two models is substantial:

$$\Delta \text{AIC} \approx 890 \quad (5.10)$$

Similarly, the difference in BIC is:

$$\Delta\text{BIC} \approx 895 \tag{5.11}$$

Differences of this magnitude provide overwhelming evidence in favor of the exponential model. This result is consistent with the theoretical expectation that muon decay follows an exponential distribution.

The Gaussian model fails to capture the strong asymmetry and long tail observed in the data, leading to a significantly poorer fit despite having an additional parameter.

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6 Results and Discussion

6.1 Summary of Results

The statistical analysis of the muon decay data yields an estimate of the lifetime parameter:

$$\hat{\tau} = 2.2078 \mu s \quad (6.1)$$

The uncertainty in the estimate has been quantified using both analytical and resampling methods. The 95% confidence interval obtained from the analytical approach is:

$$[2.0709, 2.3446] \mu s \quad (6.2)$$

The bootstrap method provides a similar interval:

$$[2.0699, 2.3417] \mu s \quad (6.3)$$

The close agreement between these intervals indicates that the estimator is stable and the uncertainty has been reliably quantified.

6.2 Goodness of Fit

The validity of the exponential model has been assessed using statistical tests.

The chi-square test yields a p-value of 0.1073, while the Kolmogorov–Smirnov test gives a p-value of 0.3496. In both cases, the p-values are sufficiently large, indicating that there is no evidence to reject the exponential model.

These results confirm that the observed data is consistent with the expected behavior of an exponential decay process.

6.3 Model Comparison

The comparison between the exponential and Gaussian models using AIC and BIC provides strong evidence in favor of the exponential model. The significantly lower values of both criteria for the exponential distribution indicate that it offers a better balance between model simplicity and goodness of fit.

This result is consistent with the theoretical expectation that muon decay follows a memoryless process, which is accurately captured by the exponential distribution.

6.4 Discussion

The analysis demonstrates that the statistical methods employed are effective in estimating the lifetime parameter and validating the underlying model. The Maximum Likelihood Estimation provides an accurate estimate, while the goodness-of-fit tests confirm the suitability of the exponential model.

The agreement between analytical and bootstrap uncertainty estimates suggests that the results are robust with respect to sampling variability. Additionally, the strong preference for the exponential model in the model comparison analysis highlights the importance of selecting physically motivated models.

It is also worth noting that the statistical methods used in this study—such as hypothesis testing, resampling techniques, and information criteria—are widely applicable to a broad range of problems in data analysis beyond this specific context.

6.5 Limitations

While the analysis provides consistent and reliable results, certain limitations should be acknowledged. The study assumes that the observations are independent and identically distributed, and that the exponential model fully captures the underlying process.

In practical experimental settings, additional factors such as detector resolution, background noise, and systematic uncertainties may introduce deviations from the ideal model. Incorporating such effects would require more sophisticated modeling and analysis techniques.

7 Conclusion

In this study, the statistical properties of muon decay times have been analyzed using a range of methods within the Frequentist framework. The lifetime parameter was estimated using Maximum Likelihood Estimation, yielding a value of $2.2078 \mu s$, which is consistent with the expected behavior of an exponential decay process.

Goodness-of-fit tests, including the chi-square test and the Kolmogorov–Smirnov test, indicate that the exponential model provides an adequate description of the data. The uncertainty in the estimated parameter was quantified using both analytical confidence intervals and bootstrap resampling, with both approaches producing consistent results.

Model comparison using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) strongly favors the exponential model over a Gaussian alternative, reinforcing the theoretical expectation for muon decay.

Overall, this work demonstrates the effectiveness of statistical inference techniques in parameter estimation, model validation, and uncertainty quantification. The consistency between different methods highlights the robustness of the analysis and illustrates the importance of combining multiple statistical tools to draw reliable conclusions from data.

The approaches used in this study are broadly applicable to a wide range of problems in physics and data analysis, where understanding both the model and the uncertainty is essential.

Code Availability

The complete implementation of the analysis, including data generation, statistical methods, and plotting routines, is available at:

<https://github.com/dineshkrishna264/muon-lifetime-statistical-analysis>

8 References

1. G. Cowan, *Statistical Data Analysis*, Oxford University Press, 1998.
2. W. T. Eadie, D. Drijard, F. E. James, M. Roos, and B. Sadoulet, *Statistical Methods in Experimental Physics*, North-Holland, 1971.
3. Particle Data Group, “Review of Particle Physics,” available at: <https://pdg.lbl.gov>
4. W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes in C*, Cambridge University Press.
5. SciPy Documentation, available at: <https://docs.scipy.org>